SOFTWARE ENGINEERING

LAB-7: FUZZING FOR INPUT VALIDATION BUGS

NAME: ANANYA

CLASS: AIML SEC 'H' 5TH SEMESTER

SRN: PES1UG24AM474

Fuzz Testing Using Hypothesis – Gym Membership Application

Original Application Execution

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>python gym_membership.py
==== Running parse_plan ====
Enter a plan code (e.g. GOLD-12): GOLD-12
Plan and Months: ('GOLD', 12)

==== Running calculate_bmi ====
Enter weight and height (e.g. 70,1.75): 70,1.75
BMI: 22.9

==== Running average_workout ====
Enter workout minutes separated by spaces (e.g. 45 60 30): 45 60 30
Average Minutes: 45
```

Initial Test Execution – Bugs Identified

The initial fuzz testing resulted in 3 failed tests, exposing multiple input validation vulnerabilities. The errors included ValueError, AttributeError, and ZeroDivisionError, caused by empty strings, invalid formats, and None values.

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>python -m pytest test_gym_membership.py -v
============================================= test session starts =============================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Ananya\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7
plugins: anyio-4.15.1, hypothesis-6.155.7, platformdirs-4.12.1
collected 3 items

test_gym_membership.py::test_parse_plan_no_crash FAILED                                                  [ 33%]
test_gym_membership.py::test_calculate_bmi_no_crash FAILED                                               [ 66%]
test_gym_membership.py::test_average_workout_no_crash FAILED                                             [100%]

================================================== FAILURES ==================================================
_____________________________________________ test_parse_plan_no_crash _____________________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 7, in test_parse_plan_no_c
rash
  |     def test_parse_plan_no_crash(s):
  |                ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 9, in test_parse_plan_no
_crash
    |     parse_plan(s)
    |     ^^^^^^^^^^^^^^
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 2, in parse_plan
    |     plan, months = code.split('-')
    |     ^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 11, in test_parse_plan_n
```

```
o_crash
    |     assert False, f"parse_plan crashed with input {s!r}: {e}"
    | AssertionError: parse_plan crashed with input '': not enough values to unpack (expected 2, got 1)
    | assert False
    | Falsifying example: test_parse_plan_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 9, in test_parse_plan_no
_crash
    |     parse_plan(s)
    |     ~~~~~~~~~~^^^
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 2, in parse_plan
    |     plan, months = code.split('-')
    |                    ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 11, in test_parse_plan_n
o_crash
    |     assert False, f"parse_plan crashed with input {s!r}: {e}"
    | AssertionError: parse_plan crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Falsifying example: test_parse_plan_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
```

```
_____ test_calculate_bmi_no_crash _____
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 15, in test_calculate_bmi_
no_crash
  |     def test_calculate_bmi_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 17, in test_calculate_bm
i_no_crash
    |     calculate_bmi(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 7, in calculate_bmi
    |     return round(float(weight) / (float(height) ** 2), 1)
    |                  ~~~~~^^^^^^^^
    | ValueError: could not convert string to float: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 19, in test_calculate_bm
i_no_crash
    |     assert False, f"calculate_bmi crashed with input {s!r}: {e}"
    | AssertionError: calculate_bmi crashed with input ',': could not convert string to float: ''
    | assert False
    | Falsifying example: test_calculate_bmi_no_crash(
    |     s=',',
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 17, in test_calculate_bm
i_no_crash
    |     calculate_bmi(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 6, in calculate_bmi
    |     weight, height = data.split(',')
```

```
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 19, in test_calculate_bm
i_no_crash
    |     assert False, f"calculate_bmi crashed with input {s!r}: {e}"
    | AssertionError: calculate_bmi crashed with input '': not enough values to unpack (expected 2, got 1)
    | assert False
    | Falsifying example: test_calculate_bmi_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 17, in test_calculate_bm
i_no_crash
    |     calculate_bmi(s)
    |     ~~~~~~~~~~~~~^^^
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 6, in calculate_bmi
    |     weight, height = data.split(',')
    |                      ^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 19, in test_calculate_bm
i_no_crash
    |     assert False, f"calculate_bmi crashed with input {s!r}: {e}"
    | AssertionError: calculate_bmi crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Falsifying example: test_calculate_bmi_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
```

```
_____________________________________ test_average_workout_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 23, in test_average_workou
t_no_crash
  |     def test_average_workout_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 25, in test_average_work
out_no_crash
    |     average_workout(s)
    |     ~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 10, in average_workout
    |     values = [int(m) for m in minutes.split()]
    |                   ~~~^^^
    | ValueError: invalid literal for int() with base 10: ':'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 27, in test_average_work
out_no_crash
    |     assert False, f"average_workout crashed with input {s!r}: {e}"
    | AssertionError: average_workout crashed with input ':': invalid literal for int() with base 10: ':'
    | assert False
    | Falsifying example: test_average_workout_no_crash(
    |     s=':',
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py:26
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 25, in test_average_work
out_no_crash
    |     average_workout(s)
```

```
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py:26
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 25, in test_average_work
out_no_crash
    |     average_workout(s)
    |     ~~~~~~~~~~~~~~~~^^^
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\gym_membership.py", line 10, in average_workout
    |     values = [int(m) for m in minutes.split()]
    |                              ^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py", line 27, in test_average_work
out_no_crash
    |     assert False, f"average_workout crashed with input {s!r}: {e}"
    | AssertionError: average_workout crashed with input None: 'NoneType' object has no attribute 'split'
    | assert False
    | Falsifying example: test_average_workout_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing examples:
    |         C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7\test_gym_membership.py:26
    +------------------------------------
========================================== short test summary info ==========================================
FAILED test_gym_membership.py::test_parse_plan_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
FAILED test_gym_membership.py::test_calculate_bmi_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
FAILED test_gym_membership.py::test_average_workout_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exception
s)
========================================== 3 failed in 8.08s ==========================================
```

Final Fuzz Testing and Test Results

After fixing the identified bugs, the application was tested again using Hypothesis and Pytest.

Initial fuzz tests after fixing – 3 passed

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>python -m pytest test_gym_membership.py -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Ananya\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7
plugins: anyio-4.15.1, hypothesis-6.155.7, platformdirs-4.12.1
collected 3 items

test_gym_membership.py::test_parse_plan_no_crash PASSED                                                                              [ 33%]
test_gym_membership.py::test_calculate_bmi_no_crash PASSED                                                                           [ 66%]
test_gym_membership.py::test_average_workout_no_crash PASSED                                                                         [100%]

============================================================ 3 passed in 0.94s ============================================================

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>
```

Strengthening the test criteria and passing

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>notepad test_gym_membership.py

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>python -m pytest test_gym_membership.py -v
============================================================ test session starts ============================================================
platform win32 -- Python 3.14.3, pytest-9.2.0, pluggy-1.7.0 -- C:\Users\Ananya\AppData\Local\Programs\Python\Python314\python.exe
cachedir: .pytest_cache
hypothesis profile 'default'
rootdir: C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7
plugins: anyio-4.15.1, hypothesis-6.155.7, platformdirs-4.12.1
collected 23 items

test_gym_membership.py::test_parse_plan_no_crash PASSED                                                                              [  4%]
test_gym_membership.py::test_calculate_bmi_no_crash PASSED                                                                           [  8%]
test_gym_membership.py::test_average_workout_no_crash PASSED                                                                         [ 13%]
test_gym_membership.py::test_parse_plan_valid PASSED                                                                                 [ 17%]
test_gym_membership.py::test_calculate_bmi_valid PASSED                                                                              [ 21%]
test_gym_membership.py::test_average_workout_valid PASSED                                                                            [ 26%]
test_gym_membership.py::test_parse_plan_invalid[None] PASSED                                                                         [ 30%]
test_gym_membership.py::test_parse_plan_invalid[] PASSED                                                                             [ 34%]
test_gym_membership.py::test_parse_plan_invalid[GOLD] PASSED                                                                         [ 39%]
test_gym_membership.py::test_parse_plan_invalid[GOLD-abc] PASSED                                                                     [ 43%]
test_gym_membership.py::test_parse_plan_invalid[-12] PASSED                                                                          [ 47%]
test_gym_membership.py::test_parse_plan_invalid[GOLD-0] PASSED                                                                       [ 52%]
test_gym_membership.py::test_calculate_bmi_invalid[None] PASSED                                                                      [ 56%]
test_gym_membership.py::test_calculate_bmi_invalid[] PASSED                                                                          [ 60%]
test_gym_membership.py::test_calculate_bmi_invalid[,] PASSED                                                                         [ 65%]
test_gym_membership.py::test_calculate_bmi_invalid[abc,1.75] PASSED                                                                  [ 69%]
test_gym_membership.py::test_calculate_bmi_invalid[70,0] PASSED                                                                      [ 73%]
test_gym_membership.py::test_calculate_bmi_invalid[70,nan] PASSED                                                                    [ 78%]
test_gym_membership.py::test_average_workout_invalid[None] PASSED                                                                    [ 82%]
test_gym_membership.py::test_average_workout_invalid[] PASSED                                                                        [ 86%]
test_gym_membership.py::test_average_workout_invalid[abc] PASSED                                                                     [ 91%]
test_gym_membership.py::test_average_workout_invalid[:] PASSED                                                                       [ 95%]
test_gym_membership.py::test_average_workout_invalid[-1 20] PASSED                                                                   [100%]

============================================================ 23 passed in 1.00s ============================================================
```

Bug Fixing and Input Validation

The application was modified to handle unexpected inputs safely.

Input type checking, format validation, numerical validation, and exception handling were implemented.

Valid Input Demonstration

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>python gym_membership.py
==== Running parse_plan ====
Enter a plan code (e.g. GOLD-12): GOLD-12
Plan and Months: ('GOLD', 12)

==== Running calculate_bmi ====
Enter weight and height (e.g. 70,1.75): 70,1.75
BMI: 22.9

==== Running average_workout ====
Enter workout minutes separated by spaces (e.g. 45 60 30): 45 60 30
Average Minutes: 45

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>
```

Invalid Input Demonstration

```
C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>python gym_membership.py
==== Running parse_plan ====
Enter a plan code (e.g. GOLD-12):
Plan and Months: Invalid input

==== Running calculate_bmi ====
Enter weight and height (e.g. 70,1.75): 5784
BMI: Invalid input

==== Running average_workout ====
Enter workout minutes separated by spaces (e.g. 45 60 30): 43 56
Average Minutes: 49

C:\Users\Ananya\OneDrive\Documents\L6_L7_21_gym_membership\Lab 7>
```

GitHub Repository

Link here